25 Solve the simultaneous equations

$$5x + y = 21$$
$$x - 3y = 9$$

5x + y = 21 (1) x - 3y = 9 (2)

Multiply all of (1) by 3, then add (2) to eliminate y.

$$\begin{array}{rrcl} & 15x + 3y & = & 63 \\ + & x - 3y & = & 9 \\ \hline & 16x & = & 72 \\ & \div 16 & & \div 16 \end{array}$$

1 mark
Opposite y coefficients: add

x = 72 ÷ 16 = 4.5

Substitute in (1): 22.5 + y = 21.

−22.5 −22.5

y = 21 − 22.5 = −1.5

1 mark
Substitute the value of x

$x =$ 4.5

$y =$ −1.5

1 mark
Both correct values

(Total for Question 25 is 3 marks)